# Maxence Debes

Data Scientist

maxence.debes@polytechnique.edu | +33 7 81 32 79 22 | Paris, France
maxencedbe.github.io | maxence-debes | maxencedbe

MSc Data Science student with a strong academic background in Mathematics and Computer Science, specialized in Machine Learning and AI. I enjoy solving complex technical problems by combining mathematical rigor with advanced Python to build robust, data-driven solutions.
Currently seeking a full-time position from October 2026.

## Education

**École Polytechnique — Polytechnic Institute of Paris** 2025 – 2026 (expected)
MSc Data Science
***Relevant coursework:*** *ALTEGRAD (joint MVA course), Deep Learning, Reinforcement Learning, Representation Learning (MVA), Generative Models, Optimization for Data Science.*

**Télécom SudParis — Polytechnic Institute of Paris** 2023 – 2026 (expected)
Master's degree in Computer Science and Applied Mathematics
***Relevant coursework:*** *Statistics and Data Analysis, Optimization, Machine Learning and Data Mining, Stochastic Processes, Continuous Optimization for Learning and Machine Learning.*

## Projects

**Hi! PARIS Hi!ckathon 2025 (2nd place)**
- Ranked 2nd out of 80+ teams in the 2025 edition of the Hi! PARIS Data Science hackathon. Task: Predicting student PISA scores from complex socio-economic backgrounds.
- Engineered a gated model routing each sample to one of two XGBoost regressors based on a feature threshold, achieving an $R^2$ of 0.79 on the public leaderboard.

**Molecular Graph Captioning (Kaggle competition)**
- Built a retrieval-based architecture using contrastive learning to align molecular graph embeddings with natural language descriptions in a joint latent space.
- Implemented a dedicated graph neural network encoder using PyTorch Geometric, optimized for semantic alignment through BERTScore and BLEU-4 metrics.

**Streaming Safety Classifier for LLMs — Capstone Project with DragonLLM (now OVHai LLM)**
- Reproduced the Qwen3Guard toxicity classifier: built a dataset of 24K toxic and safe Q&A samples, synthesized and token-level labeled with Qwen3-30B.
- Trained a lightweight per-token safety head on a frozen Qwen backbone to interrupt harmful generations in real time.

**Data Privacy & Anonymization — Project Cassiopée**
- Assessed privacy and re-identification risks in anonymized datasets through OSINT case studies, and proposed data-protection strategies. Conducted at Télécom SudParis.

**Automated Parking Optimization**
- Developed a Python optimization model for vehicle placement in multi-level automated parking, combining A* and simulated annealing to minimize retrieval costs.

## Work Experience

**Data Science Intern — ChemAI** | Munich, Germany Apr 2026 – present
- Designing the architecture of, and building, a graph database of over 30M chemical reactions.
- Running graph-native analyses to surface insights on the data's diversity, structure and reaction patterns.
- Training ML models that leverage the graph structure for predictive analytics on chemical reactions.

**Data Governance Intern — Batigère Maison Familiale** | Metz, France Jul 2025 – Aug 2025
- Migrated the company to SharePoint — adopted by 100% of employees — and designed a GDPR compliance policy reinforcing data security.

**Software Engineer Intern — Atos** | Metz, France Jul 2024 – Aug 2024
- Designed and implemented a telework tracking application using the JHipster application generator to facilitate accessibility to workstations.

## Skills & Interests

**Technical Skills:** Python, Java, SQL | PyTorch, PyG, Transformers, PEFT (LoRA/QLoRA), Scikit-learn, NumPy, SciPy, Pandas, Matplotlib | vLLM, Ollama, Unsloth, Neo4j, Weights & Biases | Git, Docker, Bash, Linux, LaTeX
**Soft Skills:** Leadership, teamwork, adaptability, technical communication
**Languages:** French (native), English (C1 – TOEIC 965/990), German (A2)
**Engagement & Interests:** Former president of a student association, competitive handball, music